

Reflecting on the entire design process, from initial brainstorming to prototyping and testing, this project has been an intense learning experience that offered both technical and interpersonal growth. Each step revealed new challenges and opportunities, ultimately shaping a stronger, more thoughtful final product. Throughout the project, our team was forced to confront real-world constraints such as limited materials, time pressure, and evolving design requirements. These obstacles, rather than hindering progress, pushed us to innovate, collaborate, and think critically.

One of the most important phases of the project was the initial research and needs assessment. At first, we underestimated the complexity of designing a containment system for a Komodo dragon. Simply containing the animal wasn't enough—we had to ensure zookeeper safety, enable efficient cleaning and animal transfers, and ensure durability in a habitat with high humidity and frequent wear. Learning to listen closely to client feedback during this stage taught us the value of understanding the true problem before rushing to solve it. We learned to ask better questions, clarify assumptions, and document requirements more thoroughly.

Design and prototyping introduced a new set of lessons. Our early ideas looked great on paper but encountered real-world failures when translated into cardboard and clamps. For example, we initially assumed that toggle clamps would hold up well, but through testing, we learned that without a stable base, the clamps could tear out of the material under high force. This forced us to rethink material choices and connection strategies. These mistakes were frustrating in the moment but ultimately made our design stronger. We also realized the importance of modularity in design—something we hadn't considered until we tried to make adjustments on the fly. Being able to disassemble and reconfigure the system became a critical asset.

The testing phase, though rushed due to time constraints, was one of the most eye-opening parts of the process. We learned that even a simple mechanism like a guillotine door requires precise tolerances and a deep understanding of friction, weight, and user interaction. The door getting stuck due to a warped panel reminded us that real materials don't always behave like ideal ones. If we had to do it again, we would build in more time for iteration and longer-term stress testing.

Working with external evaluators also highlighted the importance of communication and openness to critique. Some feedback challenged our assumptions and required us to step back and rethink key design elements. At first, it was difficult to receive criticism on something we had invested so much time into, but we gradually learned that embracing expert insight resulted in a more functional and professional outcome.

Finally, this project underscored the value of team coordination and role specialization. We learned that dividing responsibilities based on individual strengths not only improved efficiency but also allowed each team member to grow in their own area. Whether it was technical drafting, material sourcing, or field testing, everyone contributed to the success of the design.

Lessons Learned

To anyone attempting a similar project, we offer these key takeaways:

1. Define the problem deeply before designing the solution. The better you understand the user's needs and constraints, the more targeted and effective your solution will be.
2. Expect failure and plan for revision. Early prototypes are rarely perfect. Build in time for testing, failure, and iteration.

3. Prioritize communication—with your team and with your stakeholders. Regular check-ins and clear documentation can prevent misunderstandings and save time later.
4. Don't underestimate the value of expert input. Even short conversations with professionals can yield insights that dramatically improve your design.
5. Design for flexibility. Make your system adjustable and modular when possible—it will make modifications and future use much easier.

This project has shown us not just how to build a product, but how to approach a problem with curiosity, humility, and persistence. The lessons we learned extend beyond engineering; they are applicable to any collaborative, goal-driven endeavor.